

A Resonant Medium Framework for Cosmology: Finite-Memory Dynamics and the Echo Equation

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1 Introduction

Intro: context + prior art + hypothesis.

2 Datasets and Methods

Methods: data + runners + kernel fit.

3 Results

Results: Phase-8 spectral pass; Phase-20 robustness; Phase-21 guardbands.

4 Discussion

Discussion: implications, predictions, limitations.

Data/Software/Code Availability

Data/Software/Code Availability

All figures and results are reproducible using the evidence bundle in `catalog/evidence/` (archived as `echo_equation_evidence_bundle_v1.zip`). Each figure is accompanied by a provenance JSON listing its code source, input CSVs, and SHA-256 integrity hashes. The bundle also includes guard metrics and a reproducibility README for referees.

AI-use statement

We used GPT-5 Thinking (ChatGPT) for language edits, LaTeX tooling fixes, and compilation diagnostics. All conceptual content, data analysis, and results are the authors' own.

References

- [1] Planck Collaboration, *Planck 2018 results. vi. cosmological parameters*, *Astronomy & Astrophysics* **641** (2020) A6, [[arXiv:1807.06209](#)].
- [2] D. Brout and D. e. a. Scolnic, *The pantheon+ analysis: Cosmological constraints*, *Astrophysical Journal* (2022).
- [3] DESI Collaboration, *Desi 2024 bao results from dr1*, *arXiv preprint* [[arXiv:2404.03000](#)].
- [4] J. Watts, “Echo equation: Resonant medium evidence summary.” Experimentalist Lab preprint, 2025.